

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO5: AI-Based Early Detection of Alzheimer's Disease Using Medical Images. (BTL 5)

Assessment Tool Weightage: 20%

QUESTIONS: 5
Marks:50

Duration : 1 Hour
Total

Annexure A

Group assignment

Code of Conduct:

*I Jabaraj T (192511187) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:



COMMON COURSE ASSIGNMENT FORMAT

A. Assignment Information

Particular	Details
Department:	Computer Science and Engineering
Programme:	Computer Science and Engineering
Course Code & Course Name	CSA1708 & Artificial And Intelligence
Academic Year / Batch	4 th Year/2023 Batch
Faculty Name	Dr R Jeena
Assignment Title	AI-Based Early Detection of Alzheimer's Disease Using Medical Images
Date of Issue	02/09/2026
Date of Submission	02/09/2026
Maximum Marks	100
Course Outcome(s) – CO	
Bloom's Taxonomy Level	L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create
SDG Mapping (SDG 1-17)	SDG 3 – Good Health and Well-Being; SDG 9 – Industry, Innovation and Infrastructure
Industry / Societal Relevance	Healthcare AI, medical image analysis, clinical decision-support systems, early detection of neurological diseases

B. Assignment Problem / Challenge

Problem / Challenge

Alzheimer's disease is a progressive neurological disorder that affects memory, thinking, reasoning and other cognitive abilities. Early identification of Alzheimer's disease can help healthcare professionals provide appropriate care, monitoring and treatment planning.

Traditional diagnosis may involve clinical assessment, cognitive tests and examination of medical images. However, manually analyzing large numbers of medical images can be time-consuming and may be affected by human interpretation.

The proposed assignment focuses on developing an **AI-based system for early detection of Alzheimer's disease using medical images**, particularly brain MRI images. The system will use image-processing and machine-learning/deep-learning techniques to analyze brain images and classify them according to the presence or stage of Alzheimer's-related abnormalities.

The proposed system will preprocess medical images, extract meaningful features using an AI model, classify the images and evaluate the performance using appropriate metrics.

The PDF requires the assignment to be a realistic engineering/scientific/computing challenge that applies concepts, analyzes the problem, considers constraints, develops solutions and uses modern tools.

C. Problem Statement

Problem / Case / Design Challenge

"AI-Based Early Detection of Alzheimer's Disease Using Medical Images"

The objective is to design and develop an AI-based medical image analysis system capable of identifying patterns associated with Alzheimer's disease from brain medical images.

The system should accept suitable medical images such as MRI scans, perform preprocessing and normalization, extract relevant image features using an appropriate machine-learning/deep-learning model, and classify the images into suitable categories such as **Cognitively Normal, Mild Cognitive Impairment, and Alzheimer's Disease**, depending on the selected dataset.

The system should be evaluated using appropriate performance measures such as **accuracy, precision, recall, F1-score and confusion matrix**.

The system is intended as a **decision-support/research tool and not as a replacement for a qualified medical professional**.

D. Requirements and Constraints

1. Functional Requirements

1. The system should accept brain medical images as input.
2. The system should preprocess the input images.
3. The system should remove unnecessary image variations such as noise where required.
4. The system should normalize the image data.
5. The system should use an AI/deep-learning model for classification.
6. The system should generate a predicted class.
7. The system should display the prediction clearly.
8. The system should calculate performance metrics during evaluation.
9. The system should provide a confusion matrix for classification analysis.
10. The system should maintain appropriate records of experimental results.

2. Performance Requirements

1. The model should provide high classification accuracy.
2. The model should achieve good precision and recall.
3. The model should minimize false-negative predictions because missing a potentially affected case is important in medical screening.
4. The system should provide predictions within a reasonable processing time.
5. The model should generalize to previously unseen images.

3. Technical Constraints

1. The system requires an appropriate medical-image dataset.
2. Images should be in a supported format.
3. The dataset should contain sufficient samples for training and testing.
4. The AI model requires adequate computational resources.
5. Image preprocessing should be consistent across training and testing data.
6. The training and testing datasets should be properly separated to avoid data leakage.

4. Cost and Time Constraints

- The project should preferably use open-source software and publicly available research datasets where permitted.
- Development should be completed within the assigned academic period.
- Computational resources should be managed efficiently.
- Cloud computing, if used, should remain within the available budget.

5. Resource Requirements

Hardware

- Computer/laptop
- Sufficient RAM
- Storage for medical-image datasets
- GPU, if available, for deep-learning training

Software

- Python
- Jupyter Notebook / Google Colab
- TensorFlow or PyTorch
- OpenCV
- NumPy
- Pandas
- Matplotlib
- Scikit-learn

6. Safety Requirements

1. The system must not be presented as an autonomous medical diagnosis system.
2. Predictions should be treated as decision-support information.
3. Medical professionals should make the final diagnosis.
4. Incorrect predictions and model limitations should be clearly communicated.

7. Reliability Requirements

1. The system should be tested using unseen data.
2. Multiple performance metrics should be considered.
3. The model should be evaluated using appropriate validation techniques.
4. Results should be reproducible where possible.

8. Sustainability

1. Open-source tools should be preferred.
2. Computational resources should be used efficiently.
3. Lightweight models may be considered where suitable.
4. Reusable preprocessing and model pipelines should be developed.

9. Environmental Considerations

Large AI models can require significant computational power. Therefore, unnecessary training iterations should be avoided and computational resources should be used efficiently.

10. Ethical Considerations

1. Patient privacy must be protected.
2. Personal identifiers should not be exposed.
3. Medical data should only be used for permitted research/academic purposes.
4. Dataset permissions and licensing should be respected.
5. AI predictions should not be used as a substitute for professional diagnosis.
6. Possible bias in the dataset should be considered.

11. Security and Privacy

1. Medical images must be stored securely.
2. Patient-identifying information should be removed or protected.
3. Unauthorized access to medical data should be prevented.
4. Dataset and model files should be stored securely.

12. User Requirements

The system should:

- Have a simple interface.
- Allow easy image input.
- Display the prediction clearly.
- Provide understandable results.
- Display model performance.
- Avoid presenting uncertain predictions as definite medical diagnoses.

E. Student Work

1. PROBLEM UNDERSTANDING AND FORMULATION

What is the problem?

Alzheimer's disease is a progressive neurological condition that can cause memory loss and cognitive decline. Early identification can be difficult because symptoms may develop gradually.

Medical images such as brain MRI scans contain structural information that can potentially be analyzed using artificial intelligence. However, manually analyzing large image datasets is time-consuming and requires specialized expertise.

Therefore, the problem is to develop an AI-based system that can analyze medical images and identify image patterns associated with Alzheimer's disease.

Expected Outcomes

The expected outcomes are:

1. Develop a medical-image preprocessing pipeline.
2. Train an AI/deep-learning classification model.
3. Classify brain images into the selected disease categories.
4. Evaluate model performance.
5. Generate accuracy, precision, recall and F1-score.
6. Generate a confusion matrix.
7. Analyze the strengths and limitations of the model.
8. Demonstrate the feasibility of AI-assisted Alzheimer's screening.

Available Information / Data

The system can use a suitable Alzheimer's medical-image dataset containing brain MRI images and corresponding class labels.

Possible classes may include:

- Cognitively Normal
- Mild Cognitive Impairment
- Alzheimer's Disease

The exact classes depend on the selected dataset.

Assumptions

1. The medical images are correctly labeled.
2. The dataset is legally available for academic/research use.
3. Images contain sufficient information for classification.
4. Image preprocessing is applied consistently.
5. Training and testing images are appropriately separated.
6. The dataset represents the intended population reasonably well.

Constraints

1. Limited dataset size.
2. Limited computational resources.
3. Class imbalance may occur.
4. Medical images may have different resolutions.
5. AI models may produce incorrect predictions.
6. Dataset bias may affect model performance.
7. Patient privacy must be protected.
8. The system cannot replace clinical diagnosis.

2. APPLICATION OF COURSE KNOWLEDGE

A. Artificial Intelligence

Artificial intelligence enables computers to perform tasks that normally require human intelligence, such as recognizing patterns and making predictions.

In this project, AI is used to identify patterns in medical images.

B. Machine Learning

Machine learning enables the system to learn patterns from training data and make predictions on unseen data.

C. Deep Learning

Deep learning can automatically learn hierarchical image features using neural networks.

A Convolutional Neural Network (CNN) is particularly suitable for image classification because it can learn spatial patterns from images.

D. Convolution Operation

A convolution operation applies a filter/kernel over an image to extract features.

A simplified convolution can be represented as:

$$Y(i, j) = \sum_m \sum_n X(i, +mj + n)K(m, n)$$

where:

- X = input image
- K = convolution kernel
- Y = resulting feature map

E. Activation Function

A common activation function is ReLU:

$$ReLU(x) = \max(0, x)$$

It introduces non-linearity into the neural network.

F. Pooling

Pooling reduces the spatial dimensions of feature maps and helps reduce computational complexity.

For example, max pooling selects the maximum value from a region.

G. Classification

The final neural-network layer can produce class probabilities.

For multiple classes, Softmax can be used:

$$P_i = \frac{e^{z_i}}{\sum_{j=1}^N e^{z_j}}$$

where P_i represents the predicted probability for class i .

H. Loss Function

For multi-class classification, categorical cross-entropy can be used:

$$L = - \sum_i y_i \log(p_i)$$

where:

- y_i = actual class
- p_i = predicted probability

I. Performance Metrics

Accuracy

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision

$$Precision = \frac{TP}{TP + FP}$$

Recall

$$Recall = \frac{TP}{TP + FN}$$

F1-score

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

These metrics help evaluate whether the model performs reliably rather than relying only on accuracy.

3. SOLUTION / DESIGN / METHODOLOGY

Proposed System Architecture

The proposed system consists of the following stages:

Medical Image Dataset

↓

Image Preprocessing

↓

Image Resizing & Normalization

↓

Data Augmentation

↓

Training / Validation / Testing

↓

CNN / Transfer Learning Model

↓

Classification

↓

Prediction

↓

Performance Evaluation

↓

Result Analysis

Step 1: Dataset Collection

A suitable publicly available Alzheimer's brain-image dataset is selected for academic/research use.

The dataset contains medical images with corresponding labels.

Step 2: Image Preprocessing

The images are processed before being provided to the AI model.

Typical preprocessing includes:

1. Image loading
2. Resizing
3. Pixel normalization
4. Noise handling where necessary
5. Data-format conversion
6. Label encoding

For example, images can be resized to a fixed dimension such as:

224 × 224 pixels

depending on the selected model.

Step 3: Data Augmentation

To improve generalization, suitable augmentation techniques may be applied to the training images.

Examples:

- Rotation
- Zoom
- Horizontal/vertical transformations where medically appropriate
- Translation
- Brightness adjustment where appropriate

Important: Medical-image augmentation should be chosen carefully so that it does not create unrealistic anatomical patterns.

Step 4: Dataset Splitting

The dataset can be divided into:

- Training set
- Validation set
- Testing set

For example:

70% Training + 15% Validation + 15% Testing

The exact split can be adjusted according to dataset size and methodology.

Step 5: AI Model

A CNN-based architecture can be used.

CNN Pipeline

Input MRI Image

↓

Convolution Layer

↓

ReLU Activation

↓

Pooling Layer

↓

Convolution Layer

↓

ReLU

↓

Pooling

↓

Feature Extraction

↓

Fully Connected Layer

↓

Softmax Classification

Step 6: Alternative Solution

A second approach can be **Transfer Learning**.

Instead of training a CNN completely from the beginning, a pretrained image model can be adapted to the medical-image classification task.

Possible models include:

- ResNet
- DenseNet
- EfficientNet
- MobileNet

The pretrained network can be fine-tuned using the selected medical-image dataset.

Step 7: Algorithm

Proposed Algorithm

Input: Brain MRI image

Output: Predicted Alzheimer's-related class

1. Load the medical-image dataset.
2. Assign class labels.
3. Remove invalid or unusable images.
4. Resize images to the required dimensions.
5. Normalize pixel values.
6. Apply appropriate augmentation to training data.
7. Divide the dataset into training, validation and testing sets.
8. Create the CNN/transfer-learning model.
9. Train the model using training images.
10. Validate the model using validation images.
11. Tune the model if required.
12. Test the trained model using unseen test images.
13. Generate class predictions.
14. Calculate accuracy, precision, recall and F1-score.
15. Generate the confusion matrix.
16. Analyze the results.
17. Select the final model based on performance and reliability.

4. USE OF MODERN TOOLS

The PDF specifically requires evidence of appropriate tool usage and outputs.

Software Tools

Tool	Purpose
Python	Main programming language
Jupyter Notebook / Google Colab	Development and experimentation
TensorFlow / Keras	Deep-learning model development
PyTorch	Alternative deep-learning framework
OpenCV	Image processing
NumPy	Numerical computation
Pandas	Dataset/data handling
Matplotlib	Graph visualization
Scikit-learn	Metrics and evaluation
Git/GitHub	Version control and project management

Evidence of Tool Usage

The assignment should include screenshots such as:

1. Dataset loading.
2. Sample medical images.
3. Preprocessed images.
4. Training process.
5. Model architecture.
6. Training/validation accuracy graph.
7. Training/validation loss graph.
8. Confusion matrix.
9. Classification report.
10. Final prediction.

5. RESULTS AND VALIDATION

Expected Result

The developed AI model should classify the medical images into the selected Alzheimer's-related categories.

Sample Results Table

Do not present these numbers as actual results unless you obtain them from your trained model.

Metric	Obtained Value
Accuracy	[ACTUAL RESULT]
Precision	[ACTUAL RESULT]
Recall	[ACTUAL RESULT]
F1-score	[ACTUAL RESULT]

Confusion Matrix

The confusion matrix should show:

Actual / Predicted	Normal	MCI	Alzheimer's
Normal	[]	[]	[]
MCI	[]	[]	[]
Alzheimer's	[]	[]	[]

Graphs

Include:

Graph 1 – Training vs Validation Accuracy

X-axis: Epochs

Y-axis: Accuracy

Graph 2 – Training vs Validation Loss

X-axis: Epochs

Y-axis: Loss

Graph 3 – Confusion Matrix

Shows correct and incorrect classifications for each class.

Validation

The model should be validated using unseen test data.

Validation should determine:

- Whether the model satisfies the classification requirement.
- Whether the model generalizes to unseen images.
- Whether false-negative predictions are acceptably controlled.
- Whether the model performs consistently across classes.

6. ANALYSIS AND ENGINEERING DECISION

Comparison of Possible Solutions

Feature	Traditional ML	CNN	Transfer Learning
Feature extraction	Usually manual/engineered	Automatic	Automatic
Image suitability	Moderate	High	High
Training requirement	Lower	Higher	Moderate
Dataset requirement	Moderate	Large preferred	Can work well with smaller datasets
Computational requirement	Lower	Higher	Moderate–High
Complexity	Lower	High	High
Expected capability	Moderate	High	High

Selected Solution

A CNN/transfer-learning-based approach is selected because medical-image classification requires the extraction of spatial and visual features.

Transfer learning may be preferred when the available dataset is relatively limited because pretrained models can provide useful feature representations.

Advantages

1. Automated feature extraction.
2. Can process large image datasets.
3. Reduces dependence on manually designed image features.
4. Can identify complex image patterns.
5. Provides quantitative performance evaluation.

Limitations

1. Requires sufficient quality data.
2. Training can require significant computational resources.
3. Dataset bias can affect performance.
4. AI predictions can contain errors.
5. A high test accuracy does not automatically guarantee clinical usefulness.
6. Model interpretability can be challenging.

Trade-offs

A more complex deep-learning model may provide better predictive capability but may require more computational resources and can be harder to interpret.

A simpler model may be easier to train and explain but may not capture complex image patterns as effectively.

Therefore, the final model should be selected based on performance, computational requirements, reliability and interpretability, rather than accuracy alone.

7. BROADER CONSIDERATIONS

The PDF asks students to discuss sustainability, environment, society, safety, ethics, economics, accessibility and professional responsibility where relevant.

Sustainability

The use of efficient AI models and optimized training can reduce computational resource consumption.

Environment

Deep-learning training can consume considerable computational power. Efficient training and appropriate model selection can reduce unnecessary energy consumption.

Society

An AI-assisted screening system could potentially support earlier identification and improve healthcare decision-support capabilities, especially where specialist resources are limited.

Safety

The system should not independently diagnose patients. The final medical decision should remain with qualified healthcare professionals.

Ethics

Medical data is highly sensitive. Patient privacy, informed data use, fairness and responsible AI practices must be considered.

Economics

An AI-assisted screening tool may reduce some manual image-analysis workload, although proper clinical validation and deployment require significant resources.

Accessibility

A user-friendly AI system could potentially assist healthcare workers in locations with limited access to specialized medical-image analysis.

Professional Responsibility

Developers must clearly communicate:

- Model limitations.
- Expected accuracy.
- Dataset limitations.
- Possible false predictions.
- Appropriate use of the system.
- Privacy and ethical requirements.

Conclusion

The proposed AI-Based Early Detection of Alzheimer's Disease Using Medical Images system provides an AI-driven approach for analyzing brain medical images and identifying patterns associated with Alzheimer's disease. The methodology consists of image preprocessing, dataset preparation, AI-based feature extraction, classification and performance evaluation.

A CNN or transfer-learning model can be used to automatically learn relevant image features and classify the selected Alzheimer's-related categories. Performance can be evaluated using accuracy, precision, recall, F1-score and confusion matrix.

The proposed approach demonstrates the potential of artificial intelligence in medical-image analysis and decision support. However, limitations such as dataset size, class imbalance, model interpretability, generalization and data privacy must be considered.

The system should be treated as an assistive research/decision-support tool rather than a replacement for professional medical diagnosis.

Possible Improvements

1. Use larger and more diverse datasets.
2. Include multiple medical imaging modalities.
3. Apply explainable AI techniques.
4. Improve model generalization.
5. Perform external validation.
6. Investigate lightweight models.
7. Conduct clinical validation before real-world deployment.

Student Reflection

What did you learn from this assignment beyond what was directly taught in the classroom?

Through this assignment, I learned how artificial intelligence can be applied to real-world medical problems. I gained practical knowledge of medical-image preprocessing, dataset preparation, CNN-based classification, model training and performance evaluation. I also learned that high accuracy alone is not sufficient for medical applications and that privacy, ethics, reliability and model limitations are very important.

If you were given additional time/resources, what would you improve and why?

If additional time and resources were available, I would use a larger and more diverse medical-image dataset, investigate multiple deep-learning architectures and implement explainable AI techniques to understand why the model produces a particular prediction. I would also perform extensive external validation and investigate how the model could be evaluated in a realistic clinical environment.

1. References

- ☐ Alzheimer's Association, *2024 Alzheimer's Disease Facts and Figures*, Alzheimer's & Dementia, 2024.
- ☐ World Health Organization, *Dementia*, WHO.
- ☐ LeCun, Y., Bengio, Y., and Hinton, G., "Deep Learning," *Nature*, Vol. 521, 2015, pp. 436–444.
- ☐ Litjens, G., et al., "A Survey on Deep Learning in Medical Image Analysis," *Medical Image Analysis*, Vol. 42, 2017, pp. 60–88.
- ☐ Goodfellow, I., Bengio, Y., and Courville, A., *Deep Learning*, MIT Press, 2016.
- ☐ TensorFlow/Keras documentation for deep-learning model development.
- ☐ Scikit-learn documentation for machine-learning evaluation metrics.
- ☐ OpenCV documentation for image-processing operations.
- ☐ The selected Alzheimer's medical-image dataset documentation and associated publication.
- ☐ Any AI-assisted tools used during development should also be acknowledged, as specifically required by the assignment format.

F. Common Assessment Rubric

Assessment Criterion	Marks
Problem understanding & formulation	10
Application of course/domain knowledge	20
Solution methodology / design / implementation	20
Use of appropriate modern tools / techniques	10
Results, testing & validation	15
Analysis, trade-offs & justification	15
Broader considerations / professional responsibility, where applicable	5
Technical documentation & reflection	5
TOTAL	100

G. CO–PO–Assessment Mapping

Assessment Component	CO	PO(s)	Bloom's Level	Marks
Problem formulation	CO5	1,2,3,4,5,8,9,11,12	L3/L4	10
Application of knowledge	CO5	1,2,3,4,5,8,9,11,12	L3/L4	20
Solution / Design / Implementation	CO5	1,2,3,4,5,8,9,11,12	L4/L5/L6	20
Modern tool usage	CO5	1,2,3,4,5,8,9,11,12	L3/L4	10
Validation	CO5	1,2,3,4,5,8,9,11,12	L4/L5	15
Analysis & justification	CO5	1,2,3,4,5,8,9,11,12	L4/L5	15
Broader considerations	CO5	1,2,3,4,5,8,9,11,12	L4/L5	5
Documentation & reflection	CO5	1,2,3,4,5,8,9,11,12	L3/L4	5

Note: Faculty shall map only the POs and Bloom's levels genuinely assessed by the particular assignment. Every assignment is **not required to map to every PO**.

H. Faculty Evaluation & Continuous Improvement

CO Attainment

Performance Level	Criterion
Level 3	$\geq 70\%$
Level 2	60–69%
Level 1	50–59%
Level 0	$< 50\%$

Target CO Attainment: $\geq 70\%$ _____

Actual CO Attainment: **85**_____

Faculty Analysis

Areas in which students performed well: Implementation & Coding

Areas requiring improvement: Presentation

Corrective / Improvement Action:Planned to give more communication related activities

Action to be implemented in the next offering: Seminars

Documents to be Maintained

The department/course faculty shall maintain:

1. Assignment question/problem statement
2. CO–PO and Bloom's mapping
3. Assessment rubric
4. Student submissions
5. Tool/simulation/experimental evidence, where applicable
6. Evaluated scripts/reports
7. Marks and rubric-wise assessment
8. CO attainment analysis
9. Sample student work representing different performance levels
10. Faculty analysis and continuous-improvement action

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation